

# 10 V Load-Cell Excitation Rail: Ripple & Noise Characterisation

Measurement of a 5 V → 12 V boost → 10 V LDO → LC filter chain supplying strain-gauge bridge excitation and an INA118U instrumentation amplifier. Includes probe-method validation, a filter-capacitor comparison, root-cause identification, and design recommendations.

|            |                          |                                  |
|------------|--------------------------|----------------------------------|
| DATE       | INSTRUMENT               | SERIAL                           |
| 2026-08-25 | Rigo1 MH0984             | MH09B27CM00096                   |
| FIRMWARE   | DEVICE UNDER TEST        | LOAD                             |
| 00.01.00   | TPS61040DBVR boost + LDO | Strain-gauge bridge +<br>INA118U |



|          |                         |
|----------|-------------------------|
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01

Summary and verdict

The rail is **fit for purpose**. Direct-current (DC) regulation is excellent, and in the frequency band where a load cell actually operates the rail is quieter than the oscilloscope could measure. The filter is doing useful work and should be kept as originally built.

Three findings matter for the design:

|                                                                                                                                                                            |                                                                                                                                                                                                          |                                                                                                                                                                                                            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>DC OUTPUT</div> <div>9.93 V</div> <div>Drift below 1 mV over 200 ms.<br/>No control-loop instability<br/>anywhere from 10 Hz to 1 MHz.</div> <div>GOOD</div>          | <div>DOMINANT DISTURBANCE</div> <div>9.033 kHz</div> <div>A burst tone plus harmonics,<br/>inherent to the TPS61040<br/>control scheme. Filter attenuates<br/>it by 23.6 dB.</div> <div>MANAGEABLE</div> | <div>HIGHEST-RISK ITEM</div> <div>66 MHz</div> <div>Ringling at 222 mV<sub>pp</sub> on the<br/>excitation rail. Radio-frequency<br/>rectification risk at the INA118U<br/>inputs.</div> <div>ADDRESS</div> |
| <div>FILTER CAPACITOR TEST</div> <div>2.5× worse</div> <div>Substituting 0.1 µF raised<br/>wideband noise and tripled high-<br/>frequency ringing.</div> <div>REVERT</div> |                                                                                                                                                                                                          |                                                                                                                                                                                                            |

What to do

1. **Restore the original filter capacitor** and add parallel small-value ceramics (§10.1). The 0.1 µF part alone is strictly worse in every band measured.
2. **Add a radio-frequency interference (RFI) filter at the INA118U inputs** (§10.3). This is the single highest-value change, and it addresses drift rather than noise.
3. **Add or increase the feedforward capacitor on the boost converter's feedback divider** (§10.4). Texas Instruments identifies this as the direct cause of the pulse bursting we measured.
4. **Make the measurement ratiometric** (§10.6). Deriving the analogue-to-digital converter

(ADC) reference from this same rail cancels excitation ripple, the filter's DC drop, and that drop's temperature coefficient simultaneously.

CORRECTION TO NOTE

An early reading suggested a **117 mV** DC drop across the filter. That figure was an artifact of comparing two oscilloscope channels set to different vertical scales. With matched settings the difference reads **33 mV ± 50 mV** — meaning the instrument's channel-to-channel offset error exceeds the quantity being measured. **The DC drop across the filter remains unmeasured; use a digital multimeter.**

02 System under test

The power chain is:

```
5 V input
-> TPS61040DBVR boost converter -> 12 V
-> low-dropout regulator (LDO) -> 10 V [22 uF output capacitor at the LDO pins]
-> low-pass filter (LPF) -> 10 V rail
-> (a) strain-gauge bridge excitation, via connector and 2-3 ft cable
    (b) INA118U instrumentation amplifier supply
-> ADC -> robot load-cell detection
```

The significant architectural feature is that **one rail serves both the bridge excitation and the amplifier supply**. This creates two independent paths by which rail disturbance can reach the measurement, analysed in §9.

MEASUREMENT POINTS

TABLE 1

Oscilloscope channel assignment. All probes are 10:1 attenuating passive probes. "Tip-and-barrel" means the probe's ground collar is connected directly to the capacitor's ground pad with no flying ground lead.

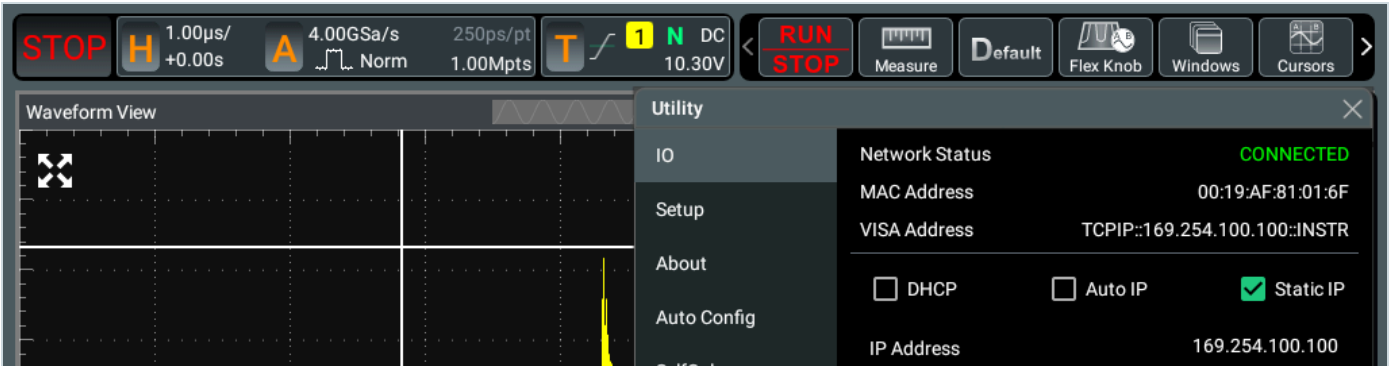
| CHANNEL | NODE                 | GROUNDING METHOD          | PURPOSE                                                       |
|---------|----------------------|---------------------------|---------------------------------------------------------------|
| CH1     | LPF output capacitor | ~15 cm flying ground clip | Deliberate negative control — shows what bad probing produces |
| CH2     | LPF output capacitor | Tip-and-barrel            | Reference measurement of the delivered rail                   |

|     |                 |                        |                                                 |
|-----|-----------------|------------------------|-------------------------------------------------|
| CH3 | LDO output pins | Direct across the pins | Filter input, for transfer-function measurement |
|-----|-----------------|------------------------|-------------------------------------------------|

### 03 Instrumentation and connection

The oscilloscope was controlled remotely over Ethernet using SCPI (Standard Commands for Programmable Instruments) over a raw transmission-control-protocol socket, which is the LXI (LAN eXtensions for Instrumentation) convention on port 5555. All waveform data in this report was transferred as binary blocks and analysed in Python; nothing was read off the screen.

| <b>TABLE 2</b><br>Network configuration. The scope address is link-local (the 169.254.x.x range), so the host adapter must sit in the same range. No manual netmask change was required – macOS auto-assigned a compatible link-local address to the USB-to-Ethernet adapter. |                              |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| PARAMETER                                                                                                                                                                                                                                                                     | VALUE                        |
| Scope IP address                                                                                                                                                                                                                                                              | 169.254.100.100              |
| Scope subnet mask                                                                                                                                                                                                                                                             | 255.255.0.0                  |
| Host adapter                                                                                                                                                                                                                                                                  | en11 (AX88179B USB-Ethernet) |
| Host IP address                                                                                                                                                                                                                                                               | 169.254.193.136/16           |
| Link speed                                                                                                                                                                                                                                                                    | 100BASE-TX full duplex       |
| SCPI socket                                                                                                                                                                                                                                                                   | 169.254.100.100:5555         |
| Round-trip latency                                                                                                                                                                                                                                                            | ~1.05 ms                     |



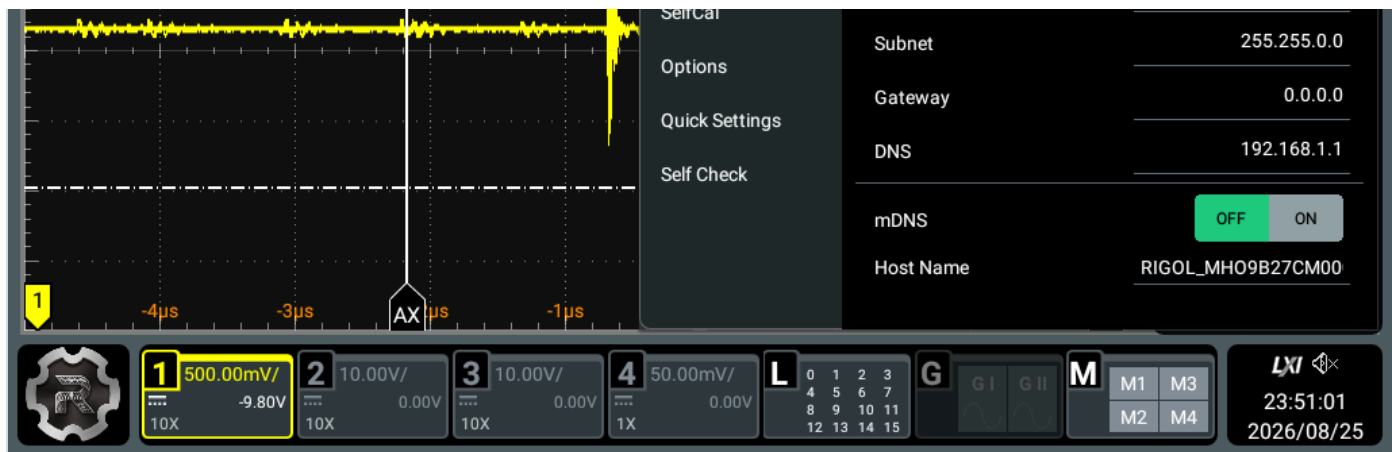


FIGURE 0

Instrument network configuration as reported by the scope's own utility page, confirming the static address and connected link. The waveform behind it is an early full-bandwidth capture of a single switching event.

#### INSTRUMENT QUIRK – WORTH KNOWING

**Alternating-current (AC) coupling on this scope settles far too slowly for low-frequency work.** After switching CH2 to AC coupling, the trace was still **9 mV** away from centre **24 seconds** later, and repeatedly drove the acquisition into saturation at sensitive vertical settings. Every low-frequency measurement in this report therefore uses **DC coupling with the vertical offset set to cancel the 9.9 V rail**. That approach settles immediately, preserves content all the way down to 0 Hz, and was verified to work at every vertical scale down to 5 mV/div.

## 04 Experimental procedure

This section is written so the measurements can be reproduced exactly. Scripts referenced here are listed in Appendix B.

### 01 Establish the remote connection

Confirm the host adapter has a link-local address in the same range as the instrument, then verify the SCPI socket responds:

```
ifconfig en11 | grep -E "status|inet "
ping -c 3 169.254.100.100
python3 -c "import socket; s=socket.create_connection(('169.254.100.100',5555),time
s.sendall(b'*IDN?\n'); print(s.recv(4096).decode())"
```

Expect `RIGOL TECHNOLOGIES,MH0984,...`. The first ping often times out while the address resolution protocol resolves; this is normal. Allow a second or two between separate socket sessions — the instrument accepts one connection at a time and refuses rapid reconnection.

02

### Probe the rail correctly

Remove the probe tip cover and use the ground spring, or solder a short wire loop from the probe's ground barrel to the capacitor's ground pad. Place the tip directly on the capacitor's positive pad. **Do not use the flying ground clip** — §5 quantifies why.

03

### Validate the probe before trusting any number

Short the probe tip to its own ground barrel and hold the pair near the converter. Capture at full bandwidth. Whatever appears is measurement artifact and sets the floor below which no reading is meaningful. Repeat this whenever the probe or its position changes.

04

### Null the DC offset instead of AC coupling

Set the channel to DC coupling and the vertical offset to the negative of the rail voltage (`:CHAN2:OFFS -9.93`). Then reduce the vertical scale until the signal occupies several divisions. This gives full sensitivity without the AC-coupling settling problem described in §3.

05

### Detect saturation explicitly

A railed acquisition returns a constant value, whose peak-to-peak spread is zero — so a clipping test written relative to the trace mean reports 0 % and silently passes. Test instead for samples pinned at the extreme quantisation code, *and* confirm the trace mean sits within the graticule. This bug produced three invalid captures before it was caught.

06

### Capture for low-frequency work

Use high-resolution acquisition mode (`:ACQ:TYPE HRES`), which averages the raw 2 GSa/s samples down to the record rate and substantially lowers the instrument noise floor. Use 20 MHz bandwidth limiting. A 1-second record at 1 MSa/s (1 Mpt memory depth) resolves from about 1 Hz to 500 kHz.

```
.venv/bin/python -c "from lfcap import lf_cap; \
lf_cap('tag', 0.1, 0.02, ch=2, coup='DC', voffs=-9.93, bwl='20M', acqtype='HRES')"
```

07

**Capture both filter sides in one acquisition**

For a transfer function the two channels must be phase-coherent, so they must come from a single acquisition rather than two separate runs:

```
.venv/bin/python -c "from capn import capN; \
capN('tag',[2,3],0.1,vdiv={2:0.05,3:0.05},voffs={2:-9.93,3:-9.96},acqtype='HRES')"
```

08

**Analyse with coherence, not just spectra**

Compute the transfer function by the cross-spectral method and **always compute coherence alongside it**. Coherence near 1 means the output genuinely derives from the input; coherence near 0 means both channels are showing their own uncorrelated noise and the transfer function there is meaningless. Without this check, ratios of two noise floors read as real filter performance.

09

**Hold the source constant across comparisons**

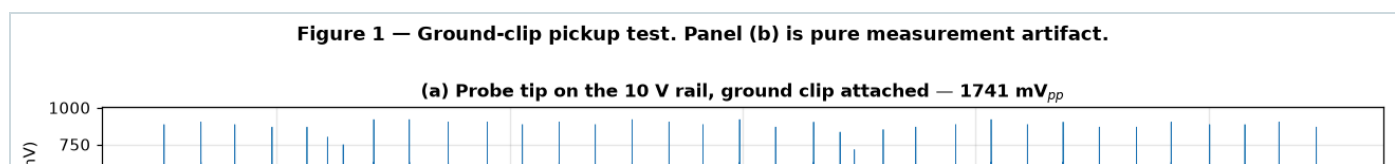
When comparing components, verify the unchanged side actually stayed unchanged. Across the capacitor swap the LDO output measured **5387  $\mu\text{V}_{\text{rms}}$**  before and **5335  $\mu\text{V}_{\text{rms}}$**  after — a 1 % difference, confirming the comparison is valid.

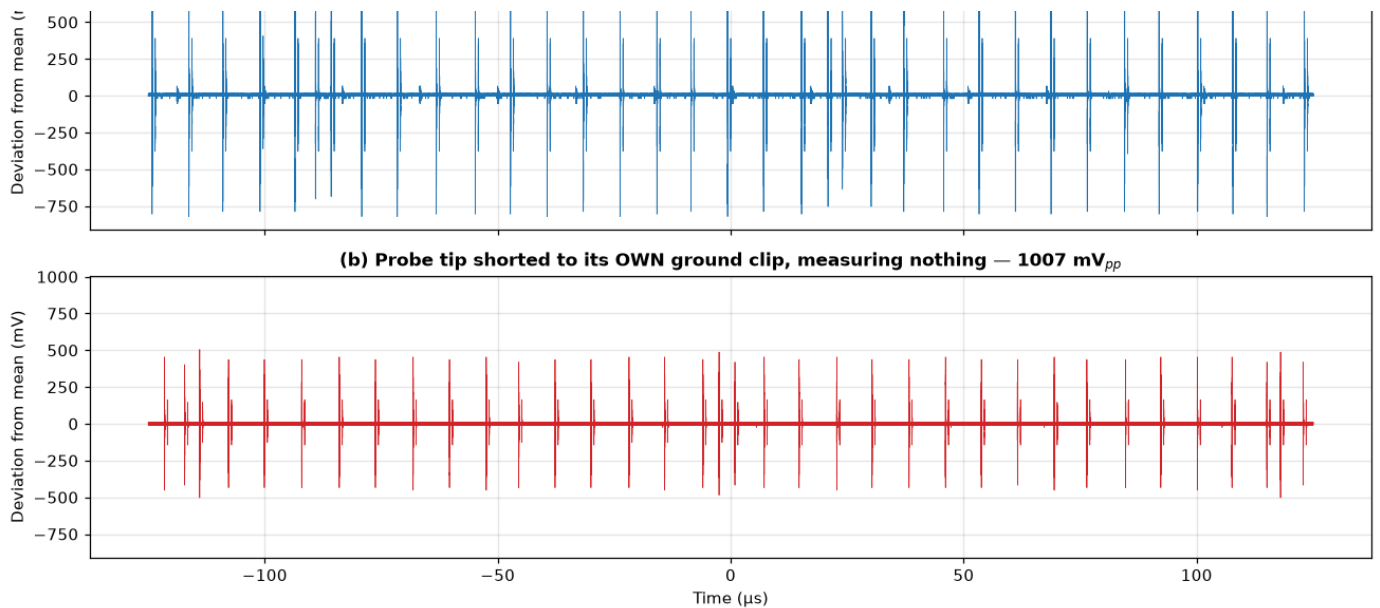
## 05 Probe validation

Before any rail measurement can be believed, the measurement method has to be shown not to manufacture the result. It very nearly did.

### 5.1 The shorted-probe test

With the probe tip shorted to its own ground clip — measuring, by construction, nothing at all — the oscilloscope still displayed **1007  $\text{mV}_{\text{pp}}$**  of spikes, at the same repetition rate as the converter.

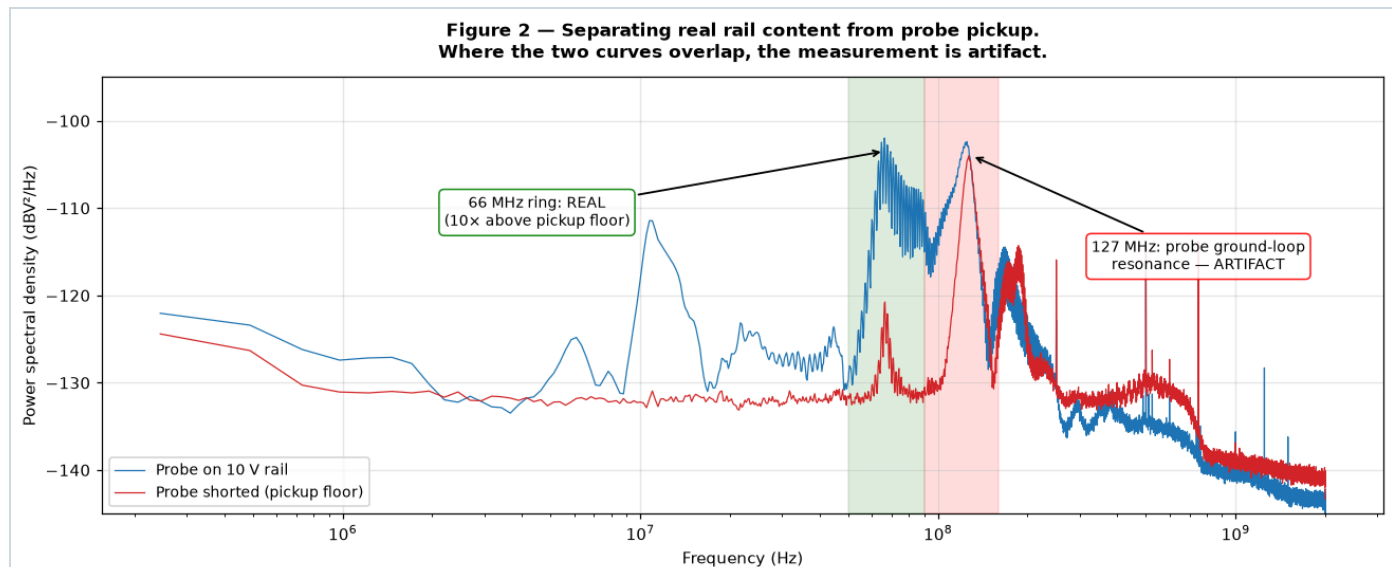


**FIGURE 1**

Ground-clip pickup test. Horizontal axis: time in microseconds. Vertical axis: deviation from the mean in millivolts, both panels on the same scale. Panel (b) is measuring nothing, so everything visible in it is loop pickup induced in the probe's own ground circuit.

## 5.2 Which content is real

Comparing the two spectra separates genuine rail content from artifact. Where the curves overlap, the reading is pickup; where the rail measurement stands clearly above the shorted-probe floor, the content is real.

**FIGURE 2**

Real content versus pickup. Horizontal axis: frequency in hertz, logarithmic. Vertical axis: power spectral density in decibels relative to 1 V<sup>2</sup> per hertz. The probe's own ground loop resonates near **127 MHz**; the **66 MHz** ring is roughly ten times above the pickup floor and is therefore genuinely present on the rail.

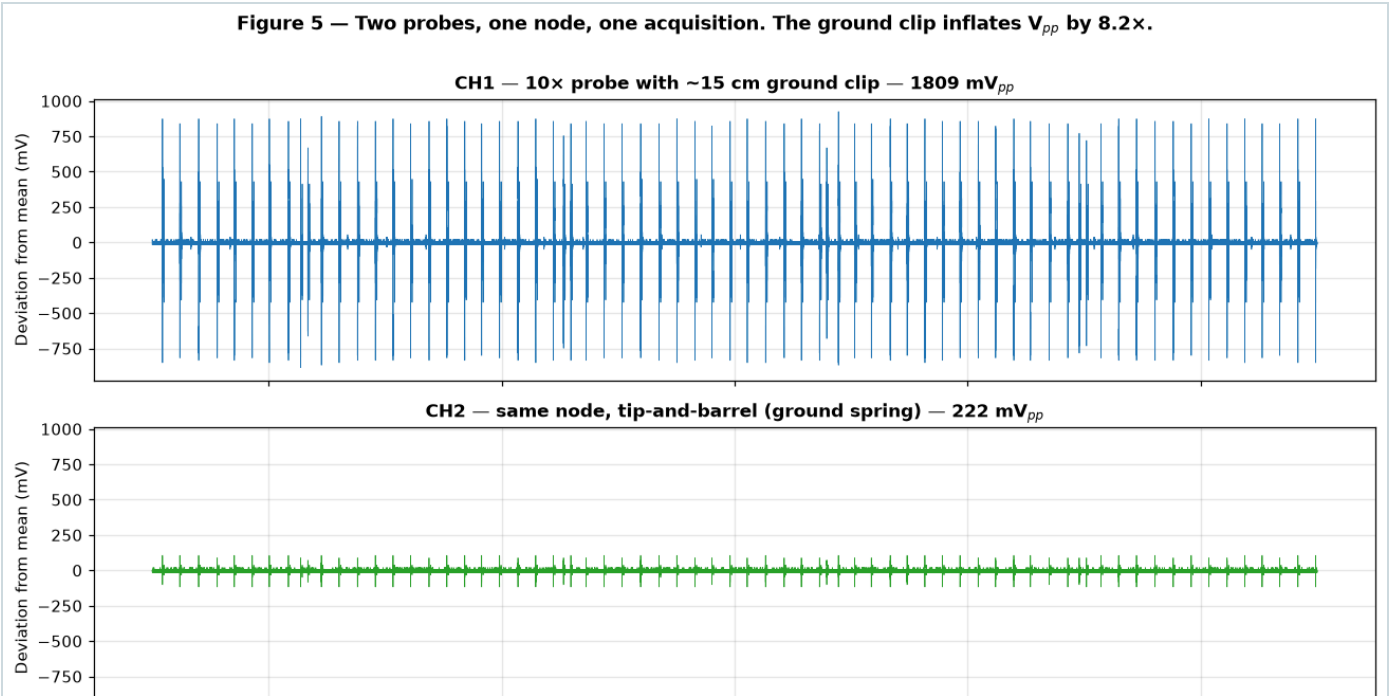
TABLE 3

Band-by-band discrimination. "Signal" is the probe on the rail; "pickup" is the shorted probe. A ratio above about 2.5 indicates real rail content; near 1 indicates the measurement is indistinguishable from artifact.

| FREQUENCY BAND | SIGNAL ( $\text{mV}_{\text{RMS}}$ ) | PICKUP ( $\text{mV}_{\text{RMS}}$ ) | RATIO | VERDICT     |
|----------------|-------------------------------------|-------------------------------------|-------|-------------|
| 0.1 – 1 MHz    | 0.639                               | 0.479                               | 1.33  | ARTIFACT    |
| 1 – 5 MHz      | 0.592                               | 0.518                               | 1.14  | ARTIFACT    |
| 5 – 20 MHz     | 4.010                               | 0.980                               | 4.09  | REAL        |
| 20 – 50 MHz    | 2.595                               | 1.388                               | 1.87  | MIXED       |
| 50 – 90 MHz    | 22.537                              | 2.285                               | 9.86  | REAL        |
| 90 – 160 MHz   | 30.463                              | 21.164                              | 1.44  | MOSTLY LOOP |
| 160 – 500 MHz  | 8.499                               | 9.581                               | 0.89  | ARTIFACT    |
| above 500 MHz  | 4.066                               | 5.791                               | 0.70  | ARTIFACT    |

5.3 Two probes, one node

With both probing methods on the same node in the same acquisition, the penalty for the flying ground clip is direct and unambiguous.



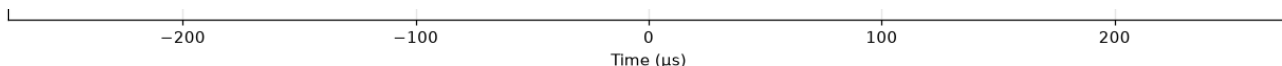


FIGURE 3

Same node, same acquisition, two grounding methods. Horizontal axis: time in microseconds. Vertical axis: deviation from the mean in millivolts, both panels on the same scale. The flying ground clip inflates peak-to-peak voltage by a factor of **8.2** and root-mean-square (RMS) voltage by a factor of **6.0**.

## METHODOLOGICAL CONCLUSION

Any switching-supply ripple figure taken with a flying ground clip should be treated as meaningless. On this rail it overstated peak-to-peak ripple by more than eight times. Every number quoted elsewhere in this report comes from tip-and-barrel probing.

## 06 Rail characterisation

## 6.1 Direct-current behaviour

TABLE 4

DC and stability measurements at the filter output.

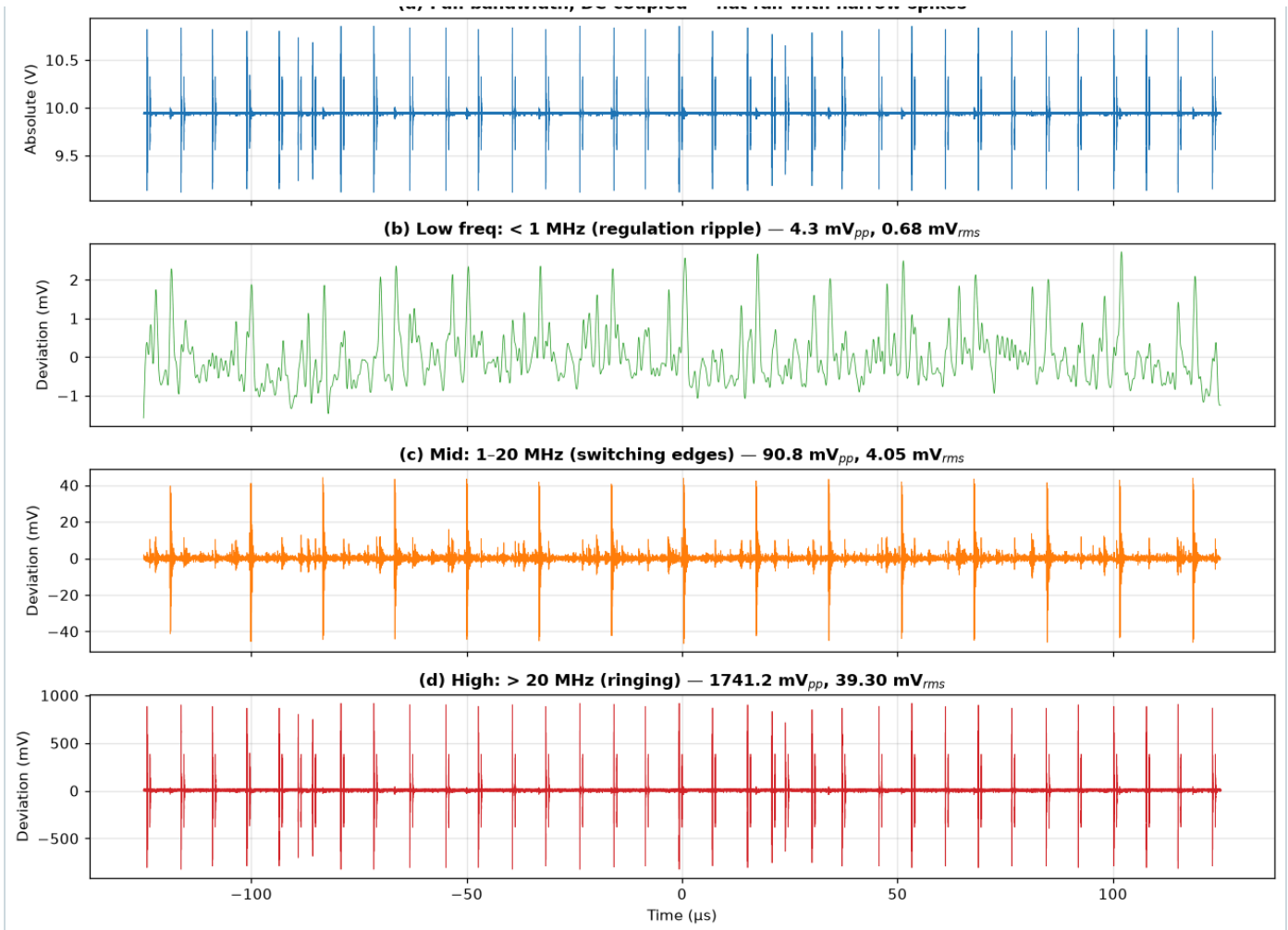
| Quantity                  | Value         | Comment                                   |
|---------------------------|---------------|-------------------------------------------|
| Output voltage            | 9.93 V        | 0.7 % below the 10 V target               |
| Drift over 200 ms         | 0.9 mV        | Excellent short-term stability            |
| Low-frequency instability | none          | No peak in the 10 Hz – 1 MHz spectrum     |
| Line-frequency ripple     | none detected | No 50/60 Hz content above the noise floor |

## 6.2 Where the ripple energy actually sits

Splitting a full-bandwidth record into frequency bands shows that the alarming raw peak-to-peak number is dominated entirely by narrow high-frequency spikes, not by regulation ripple. Between spikes the rail is a flat line.

**Figure 3 — Band decomposition of the rail (ground-clip probe). True regulation ripple is 4.3 mV<sub>pp</sub>; everything larger is high-frequency.**

(a) Full bandwidth, DC-coupled — flat rail with narrow snakes



**FIGURE 4**

Band decomposition of a 250  $\mu$ s full-bandwidth record. Horizontal axis: time in microseconds. Vertical axis: absolute rail voltage in volts for panel (a), deviation from the mean in millivolts for panels (b) to (d). Note the vertical scales differ between panels. True regulation ripple below 1 MHz is only 4.3 mV<sub>pp</sub>.

**TABLE 5**

Ripple by frequency band, from the record shown in Figure 4. Crest factor is the ratio of peak to RMS; a high value indicates narrow spikes rather than continuous ripple.

| BAND                            | V <sub>PP</sub> | V <sub>RMS</sub> | CREST FACTOR |
|---------------------------------|-----------------|------------------|--------------|
| Below 1 MHz — regulation ripple | 4.30 mV         | 0.685 mV         | 3.1          |
| 1 – 20 MHz — switching edges    | 90.83 mV        | 4.048 mV         | 11.2         |
| Above 20 MHz — ringing          | 1741 mV         | 39.301 mV        | 22.2         |
| Total, full bandwidth           | 1741 mV         | 39.533 mV        | —            |

### 6.3 The switching spike

Figure 4 — Damped ring on each switching edge, ~100 ns decay.

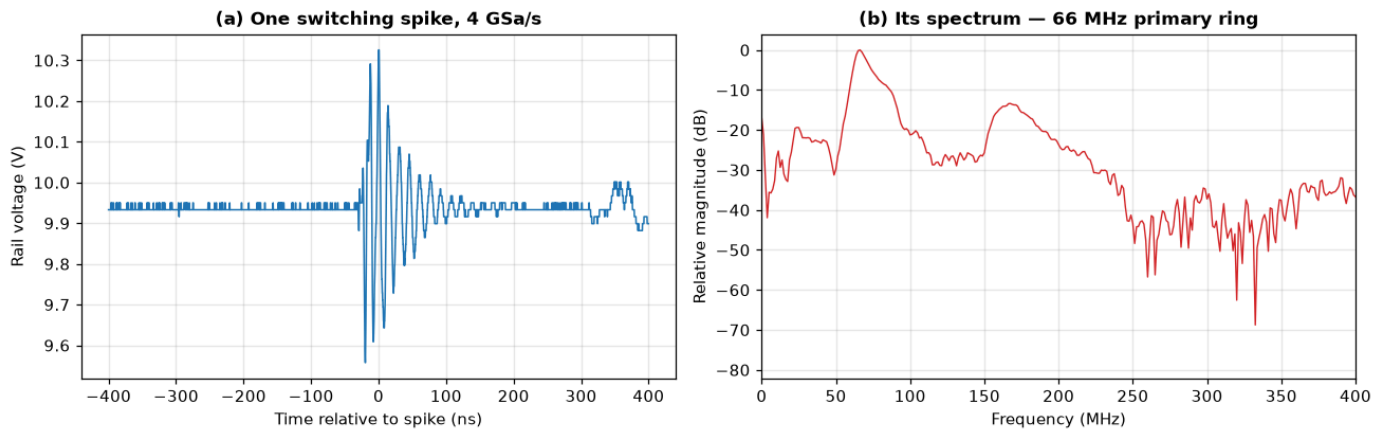


FIGURE 5

A single switching event. Left, horizontal axis: time relative to the spike in nanoseconds; vertical axis: rail voltage in volts. Right, horizontal axis: frequency in megahertz; vertical axis: magnitude in decibels relative to the peak. The ring decays in approximately 100 ns with a primary mode at **66 MHz**.

#### MEASUREMENT LIMIT

Below about 5 kHz, the filter-input and filter-output channels produce *identical* spectra with coherence under 0.05 (see Figure 6c). That means both are showing the oscilloscope's own input noise, not the rail. **The true low-frequency ripple is below the instrument's noise floor and was not measurable** even with tip-and-barrel probing, high-resolution acquisition, and offset nulling. All low-frequency figures in this report are therefore *upper bounds*. Resolving further would require a low-noise preamplifier ahead of the scope, or an AC-coupled measurement with active DC rejection.

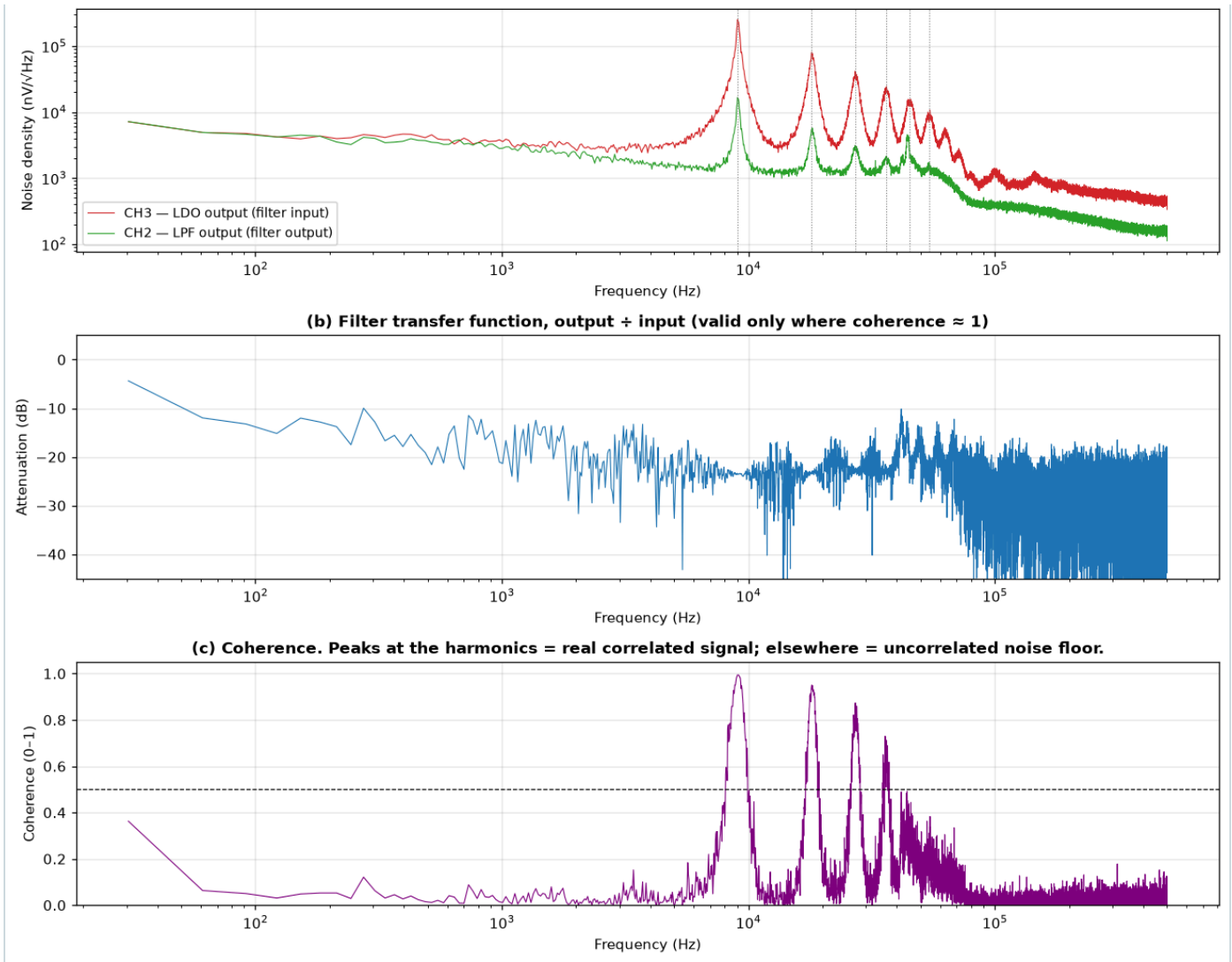
## 07 Filter characterisation

### 7.1 Baseline, original capacitor

Measuring both sides of the filter in one acquisition gives its transfer function directly. The dominant real feature is a harmonic comb whose fundamental is at **9033 Hz**, with coherence of **0.99** — unambiguously real, correlated rail content rather than noise.

Figure 6 — Baseline characterisation with the original filter capacitor.

(a) Noise spectral density each side of the filter. Dotted lines mark 9.033 kHz and harmonics.



**FIGURE 6**  
Baseline filter characterisation, original capacitor. Panel (a) horizontal axis: frequency in hertz, logarithmic; vertical axis: noise spectral density in nanovolts per root hertz. Panel (b) vertical axis: attenuation in decibels. Panel (c) vertical axis: coherence, dimensionless, 0 to 1. **Panel (c) is the key to reading panel (b)** – the transfer function is only meaningful at the frequencies where coherence approaches 1.

**TABLE 6**  
Attenuation at the burst tone and its harmonics, original capacitor. Only rows with high coherence are trustworthy.

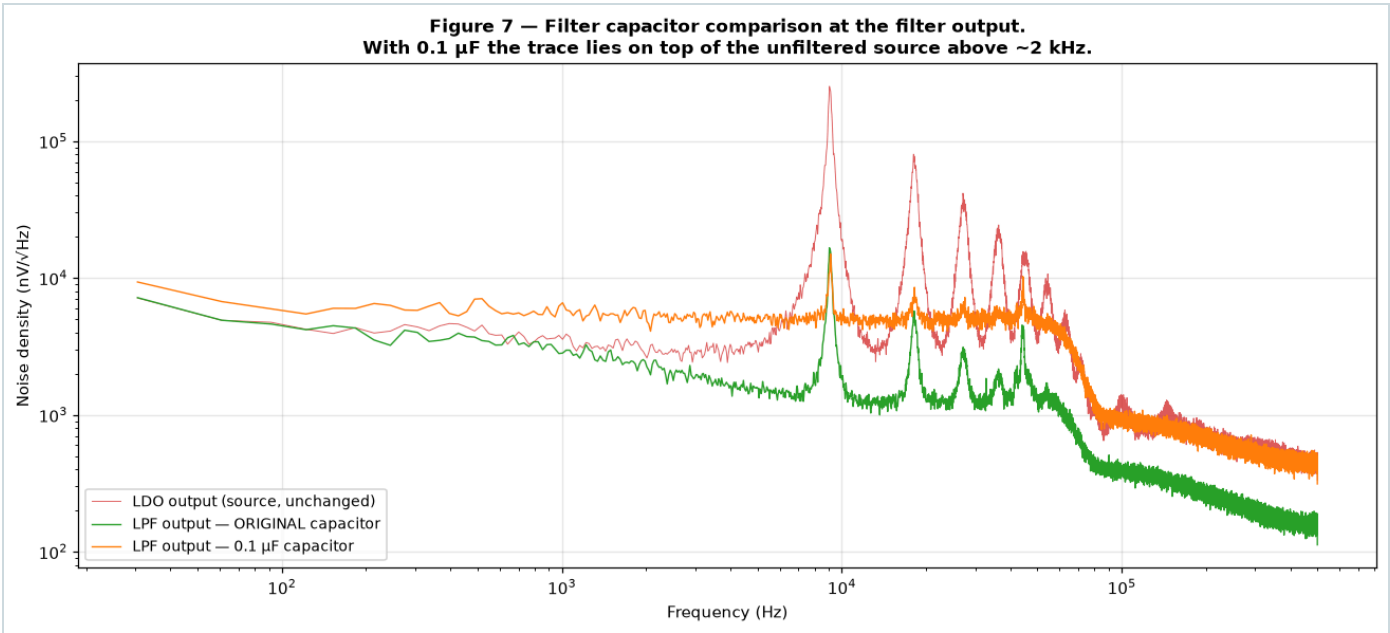
| HARMONIC    | FREQUENCY | FILTER INPUT                    | FILTER OUTPUT                   | ATTENUATION | COHERENCE |
|-------------|-----------|---------------------------------|---------------------------------|-------------|-----------|
| Fundamental | 9033 Hz   | 1391 $\mu\text{V}_{\text{rms}}$ | 91.9 $\mu\text{V}_{\text{rms}}$ | -23.6 dB    | 0.99      |
| 2nd         | 18066 Hz  | 444 $\mu\text{V}_{\text{rms}}$  | 30.4 $\mu\text{V}_{\text{rms}}$ | -23.5 dB    | 0.95      |
| 3rd         | 27100 Hz  | 230 $\mu\text{V}_{\text{rms}}$  | 17.2 $\mu\text{V}_{\text{rms}}$ | -23.2 dB    | 0.86      |
| 4th         | 36346 Hz  | 135 $\mu\text{V}_{\text{rms}}$  | 11.7 $\mu\text{V}_{\text{rms}}$ | -23.2 dB    | 0.63      |

UNEXPECTED RESULT

**The attenuation is flat, not rolling off.** A single-pole low-pass filter would give 12 dB more attenuation at 36 kHz than at 9 kHz. This one gives the same (–23.6 dB versus –23.2 dB). Combined with the capacitor-swap result below, this shows the 23 dB at 9 kHz is *not* being produced by the resistor-capacitor time constant at all. Something else — most likely the series element working against the load impedance — sets it. Identifying the mechanism requires the actual filter topology and component values, which were not available during this session.

7.2 Capacitor substitution: 0.1 µF

The 0.1 µF part was fitted for characterisation. The prediction going in was that attenuation at 9 kHz would collapse. **That prediction was wrong**, and the way it was wrong is informative: the 9 kHz attenuation barely moved, while the broadband floor between harmonics and the high-frequency content degraded substantially.

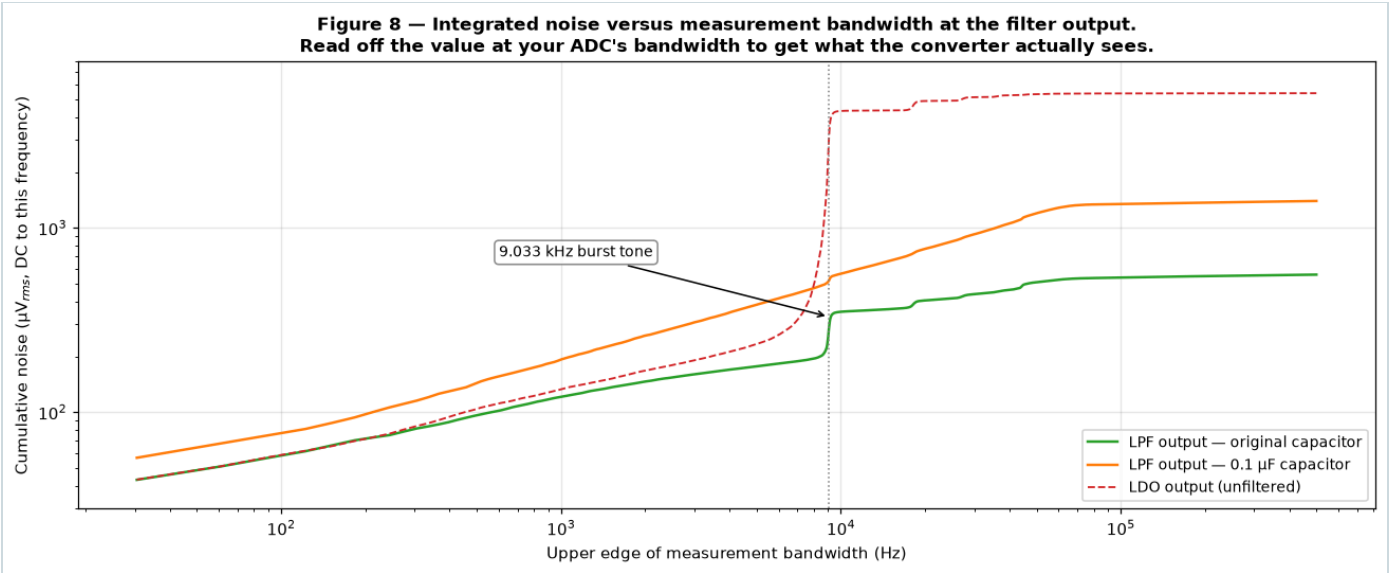


**FIGURE 7**  
Filter capacitor comparison. Horizontal axis: frequency in hertz, logarithmic. Vertical axis: noise spectral density in nanovolts per root hertz. With 0.1 µF fitted the filter output tracks the unfiltered source above roughly 2 kHz — the capacitor has effectively stopped contributing across most of the band.

**TABLE 7**  
Integrated noise at the filter output versus measurement bandwidth. Read the row matching your ADC's

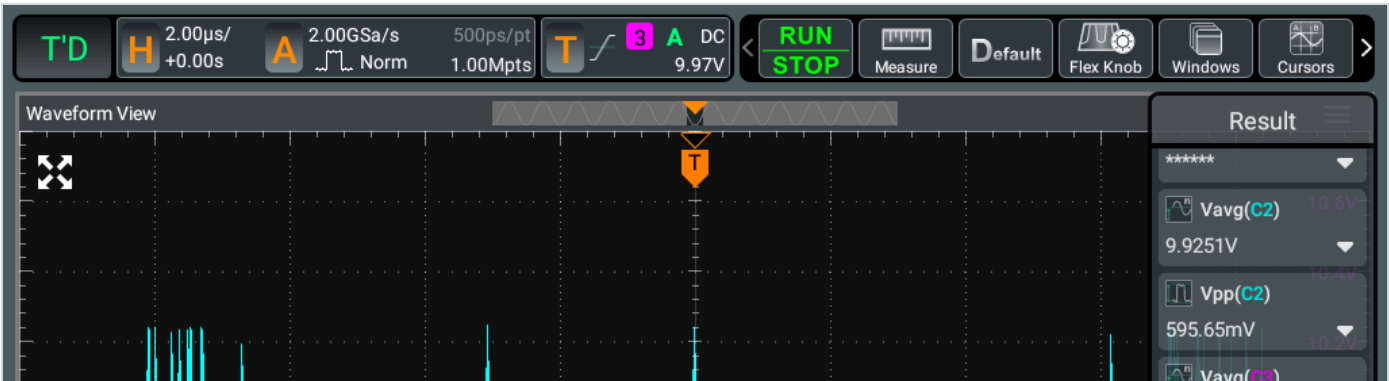
effective bandwidth. Lower is better.

| BANDWIDTH                                | ORIGINAL CAPACITOR               | 0.1 MF CAPACITOR                  | CHANGE      |
|------------------------------------------|----------------------------------|-----------------------------------|-------------|
| DC – 100 Hz                              | 56.9 $\mu\text{V}_{\text{rms}}$  | 75.3 $\mu\text{V}_{\text{rms}}$   | 1.32× worse |
| DC – 1 kHz                               | 120.3 $\mu\text{V}_{\text{rms}}$ | 190.4 $\mu\text{V}_{\text{rms}}$  | 1.58× worse |
| DC – 10 kHz                              | 350.8 $\mu\text{V}_{\text{rms}}$ | 564.2 $\mu\text{V}_{\text{rms}}$  | 1.61× worse |
| DC – 50 kHz                              | 507.7 $\mu\text{V}_{\text{rms}}$ | 1201.6 $\mu\text{V}_{\text{rms}}$ | 2.37× worse |
| DC – 500 kHz                             | 558.1 $\mu\text{V}_{\text{rms}}$ | 1400.9 $\mu\text{V}_{\text{rms}}$ | 2.51× worse |
| Above 20 MHz (ringing, $V_{\text{pp}}$ ) | 222 mV                           | 684 mV                            | 3.08× worse |



**FIGURE 8**

Integrated noise versus measurement bandwidth. Horizontal axis: upper edge of the measurement bandwidth in hertz, logarithmic. Vertical axis: cumulative noise from DC to that frequency, in microvolts RMS, logarithmic. The step at **9.033 kHz** is the burst tone; how much of it your system sees depends entirely on where your ADC's anti-alias filter sits.



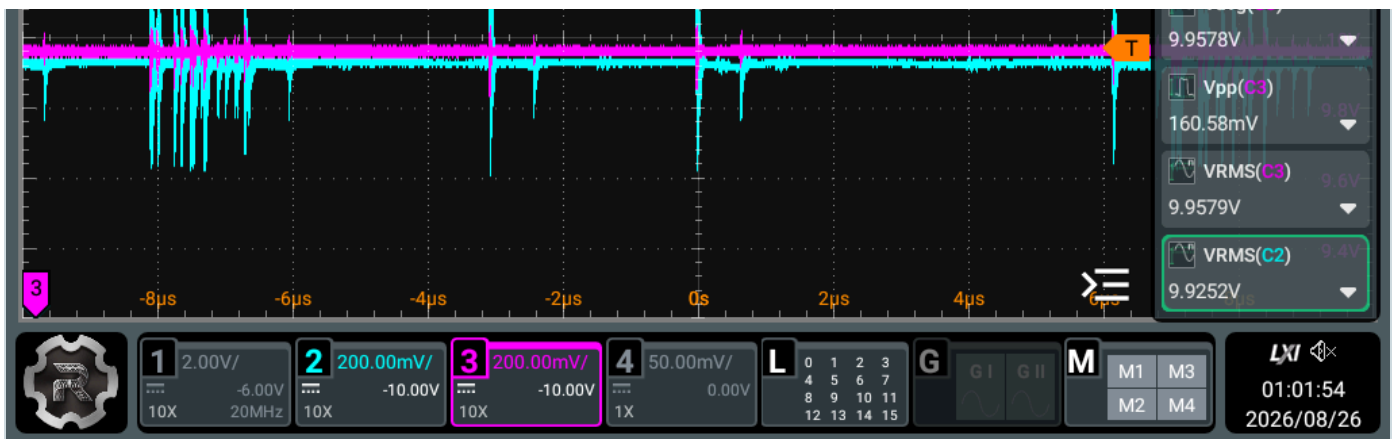


FIGURE 9

Oscilloscope capture, 2 μs/div, full acquisition bandwidth, with the 0.1 μF capacitor fitted. Magenta trace (CH3): LDO output. Cyan trace (CH2): filter output. Both at 200 mV/div. **The cluster of tightly spaced pulses on the left is the "pulse burst" behaviour described in §8.** Note also that the filter output measures 595 mV<sub>pp</sub> against the filter input's 160 mV<sub>pp</sub> — with the small capacitor fitted, the filter is making the high-frequency content *worse* than its own input.



FIGURE 10

Oscilloscope capture, 50 μs/div, 20 MHz bandwidth limit, high-resolution mode, DC-coupled with offset nulled. Magenta (CH3): LDO output at 50 mV/div. Cyan (CH2): filter output at 20 mV/div. The sawtooth on the magenta trace — droop followed by an abrupt step up — is the boost converter's hysteretic regulation cycle, which is the origin of the 9 kHz comb.

## 08 Root cause: TPS61040 control architecture

Consulting the Texas Instruments datasheet (SLVS413L) explains both the 9 kHz comb and the pulse bursting visible in Figure 9.

### 8.1 There is no fixed-frequency mode

#### DIRECT ANSWER

**No. The TPS61040 has no fixed-frequency PWM mode, and no mode pin.** The datasheet states it "operates in a pulse-frequency-modulation (PFM) scheme with constant peak current control." The part is a 5-pin device — VIN, SW, FB, EN, GND — with no mode input. The functional block diagram shows a comparator, a current-limit comparator, a 400 ns minimum-off-time block and a 6  $\mu$ s maximum-on-time block feeding a latch. Variable switching frequency is intrinsic to that architecture; it cannot be configured away.

Operation is: the switch turns on when the feedback voltage falls below the 1.233 V reference, and turns off when the inductor current reaches the current limit (400 mA typical). Each pulse therefore delivers a fixed packet of energy, and the *rate* of pulses varies with load. The comb at 9.033 kHz is the repetition rate of the regulation cycle: the converter delivers a burst of pulses, the output rises, the comparator stops switching, the output droops, and the cycle repeats. That is exactly the sawtooth visible on the magenta trace in Figure 10.

### 8.2 The pulse bursting is a documented, fixable condition

The datasheet is explicit about what causes the burst pattern seen in Figure 9:

#### DATASHEET, SECTION 7.2.2.2

"A feedforward capacitor across the upper feedback resistor R1 **is required** to provide sufficient overdrive for the error comparator. Without a feedforward capacitor, or one whose value is too small, the TPS6104x shows **double pulses or a pulse burst** instead of single pulses at the switch node, causing higher output voltage ripple."

The datasheet also caveats its own output-ripple equation with the phrase "assuming the converter does not show double pulses or pulse bursts on the switch node" — that is, the standard ripple calculation does not apply to a converter in the state we measured.

The recommended feedforward capacitance is

$$C_{FF} = 20 / (2 * \pi * f_S * R1)$$

where `f_S` is the switching frequency at the nominal load and `R1` is the upper feedback-divider resistor. The datasheet's own worked example — 18 V output, R1 = 2.2 MΩ — arrives at 22 pF. A 10 pF starting point is suggested for general use, with larger values needed at lower switching frequencies.

TRADE-OFF

The datasheet notes that a larger feedforward capacitor worsens line regulation. For this application, where the input is a regulated 5 V rather than a discharging battery, line regulation is a minor concern and ripple reduction is the priority.

8.3 Relevant device parameters

TABLE 8

TPS61040 parameters from datasheet SLVS413L, for reference when recalculating the feedforward capacitor or output capacitance.

| PARAMETER                    | MIN    | TYPICAL | MAX    |
|------------------------------|--------|---------|--------|
| Switch current limit         | 350 mA | 400 mA  | 450 mA |
| Minimum off time             | 250 ns | 400 ns  | 550 ns |
| Maximum on time              | 4 μs   | 6 μs    | 7.5 μs |
| Switching frequency          | —      | —       | 1 MHz  |
| Feedback reference           | —      | 1.233 V | —      |
| Recommended inductor         | 2.2 μH | —       | 10 μH  |
| Recommended output capacitor | 1 μF   | —       | —      |

Measured switching behaviour was consistent with these figures: individual pulses within a burst were spaced roughly 7 μs apart (about 128 kHz), with the on-time appearing as paired edges separated by roughly 0.67 μs.

09

Impact on the load-cell signal chain

Because the 10 V rail feeds both the bridge excitation and the INA118U supply, rail disturbance reaches the measurement by three distinct paths. Quantifying each shows which ones matter.

9.1 Path analysis

Assumptions: a 2 mV/V load cell, full-scale load, gain of 100, and the measured 9 kHz tone of **91.9  $\mu\text{V}_{\text{rms}}$**  at the filter output. Pessimistic rejection figures are used deliberately.

TABLE 9

Rail disturbance referred to the amplifier input. "Referred to input" means the equivalent signal at the bridge output that would produce the same effect, which is the fair basis for comparison against amplifier noise and load-cell resolution.

| PATH                     | MECHANISM                                                                       | REJECTION ASSUMED | REFERRED TO INPUT      |
|--------------------------|---------------------------------------------------------------------------------|-------------------|------------------------|
| Excitation, differential | Bridge output scales with excitation, multiplied by cell sensitivity (0.002)    | 54 dB (inherent)  | 184 nV <sub>rms</sub>  |
| Excitation, common mode  | Bridge midpoints move by half the excitation ripple; rejected by amplifier CMRR | 60 dB             | 46 nV <sub>rms</sub>   |
| Amplifier supply         | Rejected by amplifier PSRR                                                      | 60 dB             | 92 nV <sub>rms</sub>   |
| Total (root-sum-square)  | —                                                                               | —                 | ≈210 nV <sub>rms</sub> |

ASSESSMENT

Roughly **210 nV<sub>rms</sub>** referred to the input is comparable to the INA118U's own input noise over a bandwidth of a few hundred hertz, and it sits at 9 kHz where any reasonable anti-alias filter removes it. Against a 20 mV full-scale signal this corresponds to roughly **12,000 noise-limited counts** before anything else in the chain contributes. **The rail is not the limiting element in this measurement chain** — the load cell and the amplifier will be.

Note that the excitation path is *self-attenuating*: a Wheatstone bridge converts excitation ripple to differential output only in proportion to its own imbalance. At zero load the bridge is balanced and excitation ripple appears purely as common mode. This is why the excitation path, despite having no active rejection, contributes only modestly.

## 9.2 The aliasing caveat

The 9 kHz tone is harmless *if it is filtered before sampling*. If it is not, it folds into the measurement band and appears as slow wander in the load reading that no amount of averaging will remove. With a sampling rate of 1 kSPS, for instance, 9033 Hz aliases to 33 Hz. A simple resistor-capacitor anti-alias filter at a few hundred hertz eliminates the concern entirely.

## 9.3 The high-frequency risk is different in kind

HIGHEST-PRIORITY FINDING

The rail carries 222 mV<sub>pp</sub> of ringing above 20 MHz, confirmed real in §5.2. **The INA118U has no input radio-frequency filtering.** Instrumentation amplifiers rectify high-frequency energy at their input stage and convert it into a *DC offset that drifts* — which, unlike noise, cannot be averaged away and reads as phantom force. A 2–3 ft cable to the load cell acts as an efficient antenna at these frequencies in both directions. This is the one finding that warrants a board change rather than a tuning adjustment.

# 10 Recommendations

## 10.1 Filter capacitor for the final design

**Restore the original capacitor and add small-value ceramics in parallel.** The 0.1 µF part was worse in every band measured (Table 7). The reason a single 0.1 µF fails is that it has no bulk capacity; the reason a single large capacitor is also imperfect is that its equivalent series inductance (ESL) makes it ineffective at high frequency. A parallel combination covers both.

|                                                                                                                                              |
|----------------------------------------------------------------------------------------------------------------------------------------------|
| TABLE 10                                                                                                                                     |
| Recommended filter-output capacitor bank. Place all three physically close to the filter output, with the smallest package nearest the load. |

| ROLE           | VALUE      | DIELECTRIC / PACKAGE | PURPOSE                                             |
|----------------|------------|----------------------|-----------------------------------------------------|
| Bulk           | 22 $\mu$ F | X7R, 25 V, 1210      | Dominates from below 1 kHz to ~100 kHz              |
| Mid            | 1 $\mu$ F  | X7R, 25 V, 0603      | Covers the gap where the bulk part's ESL takes over |
| High frequency | 100 nF     | X7R, 25 V, 0402      | Low ESL; effective into the tens of megahertz       |

DO NOT SKIP THIS – DC BIAS DERATING

Class II ceramic capacitors lose a large fraction of their capacitance under DC bias. A 22  $\mu$ F X7R rated at 16 V can fall to **7–10  $\mu$ F at 10 V** of applied bias. Specify **25 V or 50 V rating and the largest package you can afford**, and check the manufacturer's bias-derating curve rather than the nameplate value. This effect is large enough to explain a filter underperforming its calculated design.

10.2 Ferrite bead and ceramic on the excitation output

The target is the 66 MHz ring, so the bead must present high impedance in the tens-to-hundreds of megahertz range. The binding constraint is **DC resistance**: excitation current flows through this bead, and its resistance becomes a load-dependent, temperature-dependent voltage drop directly in your excitation path.

TABLE 11

Ferrite bead selection criteria. Select on these parameters rather than on a specific part number; verify against current stock. A 350  $\Omega$  bridge at 10 V draws roughly 29 mA.

| PARAMETER            | TARGET             | REASONING                                                           |
|----------------------|--------------------|---------------------------------------------------------------------|
| Impedance at 100 MHz | 120 – 300 $\Omega$ | Covers the measured 66 MHz ring and its second mode near 170 MHz    |
| DC resistance        | below 0.1 $\Omega$ | At 29 mA this keeps the excitation drop under 3 mV                  |
| Rated current        | $\geq$ 1 A         | Well above the operating point, so the bead stays out of saturation |
| Package              | 0805 or 1206       | Larger packages give lower DC resistance for the same impedance     |

A representative part meeting these criteria is the **Murata BLM21PG221SN1D** (0805, 220  $\Omega$  at 100 MHz, 2 A, 50 m $\Omega$  DC resistance) — confirm the current specification and availability before committing. Follow the bead immediately with **100 nF X7R 0402** and **1  $\mu$ F X7R 0603 to ground**.

CHECK FOR PEAKING

A ferrite bead behaves as an inductor below its resonance, so a bead-plus-capacitor network can *amplify* signals at its own resonant frequency — typically somewhere in the 100 kHz to 1 MHz range, which is where your switching harmonics live. After fitting, re-measure to confirm no new peak has appeared. If one has, damp it with a few ohms in series or increase the capacitance.

10.3 RFI filter at the INA118U inputs

This is the highest-value change in this report. It is the standard Burr-Brown input network:

TABLE 12

Input RFI filter component values. The differential capacitor must be at least ten times the common-mode capacitors, so that mismatch between the two common-mode parts — which converts common-mode signal into differential error and degrades CMRR — is swamped.

| COMPONENT                  | VALUE        | PLACEMENT                | NOTES                                                     |
|----------------------------|--------------|--------------------------|-----------------------------------------------------------|
| Series resistors (2)       | 1 k $\Omega$ | One in each input leg    | <b>Match to 0.1 %</b> — mismatch directly degrades CMRR   |
| Common-mode capacitors (2) | 1 nF         | Each input leg to ground | COG/NP0 dielectric for stability                          |
| Differential capacitor (1) | 10 nF        | Between the two legs     | COG or X7R; must be $\geq 10\times$ the common-mode value |

Resulting bandwidths: common mode approximately **159 kHz**, differential approximately **3.8 kHz**. The differential figure is generous for a load cell and can be reduced further by increasing the differential capacitor if you want additional anti-alias margin.

Cost: the 1 k $\Omega$  resistors add roughly **5.7 nV/ $\sqrt{\text{Hz}}$**  of thermal noise differentially, against the INA118U's own input noise of about 10 nV/ $\sqrt{\text{Hz}}$  — an increase of approximately 16 % in total input noise. If that matters more to you than RF rejection, drop to 470  $\Omega$  and double the capacitors.

## 10.4 Boost converter changes

1. **Fit or increase the feedforward capacitor** across the upper feedback resistor on the TPS61040 board (§8.2). This directly targets the pulse bursting and is Texas Instruments' own prescribed remedy. Start at 22 pF if the divider resistance is in the megohm range.
2. **Increase the boost output capacitance** on the 12 V rail. The datasheet states plainly that "increasing the output capacitor value reduces the output voltage ripple" in exactly this bursting regime. This is the easiest high-leverage change if the feedback network is not accessible on the module.
3. **If neither is possible**, accept the 9 kHz comb and rely on the LDO, the filter, and the ADC anti-alias filter. §9 shows this is tolerable — it is simply not optimal.

## 10.5 Where to place the filtering: PCB or load-cell end?

DIRECT ANSWER

Put everything on your PCB. Nothing needs to go at the load-cell end of the cable. The two placements do different jobs, and both jobs are done at the board.

| TABLE 13                                                                                                                                                                    |                                            |                                                                                                                                                          |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Component placement. The governing principle: filter the excitation <i>before</i> it reaches the cable, and filter the returning sense lines <i>at the amplifier pins</i> . |                                            |                                                                                                                                                          |
| COMPONENT                                                                                                                                                                   | WHERE                                      | WHY THERE                                                                                                                                                |
| Ferrite bead + ceramics on excitation                                                                                                                                       | PCB, immediately at the connector          | Stops high-frequency energy getting onto the cable at all. Filtering at the far end is too late — the cable has already radiated along its whole length. |
| RFI filter on sense lines                                                                                                                                                   | PCB, immediately at the INA118U input pins | Rectification happens at the amplifier's input stage. The RF arrives via the cable, so the filter must sit between the connector and the pins.           |
| Filter capacitor bank                                                                                                                                                       | PCB, at the filter output                  | Local decoupling only works when it is local.                                                                                                            |
| Anything at the load cell                                                                                                                                                   | <b>Not required</b>                        | The bridge is a low-impedance resistive network and does not rectify. Load cells are usually sealed with moulded cable, making this impractical anyway.  |

10.6 Cable and architecture – larger wins than component values

1. **Use six-wire (Kelvin) sensing if the load cell supports it.** Dedicated sense lines carry no current, so every resistive drop – connector, cable, ferrite bead DC resistance, and the filter's series element – cancels completely. This single change makes the unresolved DC-drop question in §1 irrelevant.
2. **Make the ADC reference ratiometric** by deriving it from the same 10 V rail. Excitation ripple, the filter's DC drop, and that drop's temperature coefficient all cancel to first order. This is the standard load-cell architecture and it neutralises several findings at once.
3. **Use shielded cable with twisted pairs** – excitation positive with excitation negative, sense positive with sense negative. Ground the shield *at the PCB end only* to avoid a ground loop.
4. **Consider a common-mode choke** on the whole cable bundle at the connector. On a robot with motors and speed controllers nearby, this is worth more than any single-ended filter.
5. **Route the load-cell cable away from motor and controller wiring.** Physical separation is free and outperforms components.

11 Open items

| <div>TABLE 14</div> <div>Questions this session could not close, with the measurement that would close each.</div> |                                                                                                                                                       |
|--------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| OPEN QUESTION                                                                                                      | HOW TO RESOLVE IT                                                                                                                                     |
| DC drop across the filter                                                                                          | Digital multimeter across the filter's series element. The oscilloscope's channel-to-channel offset error ( $\pm 50$ mV) exceeds the quantity.        |
| What actually sets the flat 23 dB at 9 kHz                                                                         | Provide the filter topology and component values. Changing capacitance by two orders of magnitude did not move it, so it is not the RC time constant. |
| True low-frequency ripple                                                                                          | Low-noise preamplifier ahead of the scope. Current figures are upper bounds set by the instrument, not the rail.                                      |
| Boost output (12 V) ripple, and hence LDO rejection                                                                | Probe the 12 V rail on CH1 with tip-and-barrel grounding. This would give the LDO's actual PSRR at 9 kHz.                                             |

8/25/26, 1:15 PM

Load Cell Rail Characterisation

|                                                     |                                                                                                                                            |
|-----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Whether the module's feedback network is accessible | Inspect the Adafruit board for the divider resistors and any existing feedforward capacitor.                                               |
| Load current at the operating point                 | Needed to confirm the TPS61040 has adequate margin. At 5 V in and 12 V out with a 400 mA switch limit, available output current is modest. |

A

# Glossary

|                   |                                                                                                                                                                                                                          |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AC                | Alternating current. In oscilloscope context, "AC coupling" inserts a capacitor to block the DC component of a signal.                                                                                                   |
| ADC               | Analogue-to-digital converter. The device that samples the amplified bridge signal into digital form.                                                                                                                    |
| Anti-alias filter | A low-pass filter placed before a sampler, to prevent signals above half the sampling rate from folding down into the measurement band as false low-frequency content.                                                   |
| BWL               | Bandwidth limit. An oscilloscope setting that inserts a low-pass filter, here 20 MHz, ahead of the acquisition.                                                                                                          |
| CMRR              | Common-mode rejection ratio. How well a differential amplifier ignores a signal appearing identically on both inputs. Higher is better.                                                                                  |
| Coherence         | A dimensionless number from 0 to 1 measuring how much of one signal is linearly derived from another at a given frequency. Near 1, a measured transfer function is meaningful; near 0, it is a ratio of unrelated noise. |
| Crest factor      | Ratio of peak amplitude to RMS amplitude. A high value indicates narrow spikes rather than continuous ripple.                                                                                                            |
| DC                | Direct current. The steady, non-varying component of a voltage.                                                                                                                                                          |
| DCR               | DC resistance. The resistance a component presents to steady current — for a ferrite bead, the source of unwanted voltage drop.                                                                                          |
| ESL               | Equivalent series inductance. Parasitic inductance in a real capacitor, which makes it ineffective above a certain frequency.                                                                                            |
| ESR               | Equivalent series resistance. Parasitic resistance in a real capacitor, which limits how far it can suppress ripple.                                                                                                     |

|                       |                                                                                                                                                                                                                             |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>FFT</b>            | Fast Fourier transform. The algorithm used to convert a time-domain record into a frequency spectrum.                                                                                                                       |
| <b>HRES</b>           | High-resolution acquisition mode. The oscilloscope averages many raw samples per stored point, trading bandwidth for a lower noise floor and more effective bits.                                                           |
| <b>INA118U</b>        | The instrumentation amplifier in this design — a precision differential amplifier intended for bridge sensors.                                                                                                              |
| <b>Kelvin sensing</b> | Also "four-wire" or, for a bridge, "six-wire" sensing. Separate force and sense connections, so that voltage drops in the current-carrying wires do not appear in the measurement.                                          |
| <b>LDO</b>            | Low-dropout regulator. A linear regulator that produces a clean output from a slightly higher, noisier input.                                                                                                               |
| <b>LPF</b>            | Low-pass filter. Passes low frequencies, attenuates high ones.                                                                                                                                                              |
| <b>LXI</b>            | LAN eXtensions for Instrumentation. The Ethernet-based standard by which the oscilloscope was controlled.                                                                                                                   |
| <b>PFM</b>            | Pulse-frequency modulation. A switching-converter control scheme in which pulse size is fixed and pulse <i>rate</i> varies with load. Efficient at light load, but produces a load-dependent, variable switching frequency. |
| <b>PSD</b>            | Power spectral density. How signal power is distributed across frequency, in units of volts squared per hertz.                                                                                                              |
| <b>PSRR</b>           | Power-supply rejection ratio. How well an amplifier ignores disturbance on its supply rail. Typically excellent at DC and progressively worse with frequency.                                                               |
| <b>PWM</b>            | Pulse-width modulation. A control scheme in which switching frequency is fixed and pulse <i>width</i> varies. Produces predictable, easily filtered ripple — the TPS61040 does not offer it.                                |
| <b>RFI</b>            | Radio-frequency interference. High-frequency energy that can be rectified by an amplifier's input stage into a DC offset error.                                                                                             |
| <b>RMS</b>            | Root mean square. The effective average magnitude of a varying signal — the appropriate measure for noise.                                                                                                                  |
| <b>SCPI</b>           | Standard Commands for Programmable Instruments. The text command language used to control the oscilloscope remotely.                                                                                                        |
| <b>Tip-and-barrel</b> | A probing method in which the probe's ground barrel connects directly to the target's ground pad, eliminating the flying ground lead and the pickup                                                                         |

loop it forms.

$V_{pp}$

Peak-to-peak voltage — the full span from most negative to most positive excursion.

Wheatstone bridge

The four-resistor network inside a load cell. Its output is proportional to both applied strain and excitation voltage.

B Files and scripts

ANALYSIS CODE

TABLE 15  
Scripts written during this session, in the project directory.

| FILE                        | PURPOSE                                                                                                   |
|-----------------------------|-----------------------------------------------------------------------------------------------------------|
| scope.py                    | SCPI client: connection with retry, IEEE 488.2 block transfer, paged waveform capture, screenshot capture |
| capn.py                     | Phase-coherent multi-channel capture from a single acquisition, with saturation detection                 |
| lfcap.py                    | Long-record, high-resolution, offset-nulled low-frequency capture                                         |
| dual.py , zoom.py , grab.py | Earlier single- and dual-channel capture helpers                                                          |
| makefigs.py                 | Generates every figure in this report from the saved data                                                 |

RAW DATA